Turning large language models into cognitive models

Marcel Binz
MPRG Computational Principles of Intelligence
Max Planck Institute for Biological Cybernetics, Tübingen, Germany
marcel.binz@tue.mpg.de

Eric Schulz
MPRG Computational Principles of Intelligence
Max Planck Institute for Biological Cybernetics, Tübingen, Germany

Abstract

Large language models are powerful systems that excel at many tasks, ranging from translation to mathematical reasoning. Yet, at the same time, these models often show unhuman-like characteristics. In the present paper, we address this gap and ask whether large language models can be turned into cognitive models. We find that – after finetuning them on data from psychological experiments – these models offer accurate representations of human behavior, even outperforming traditional cognitive models in two decision-making domains. In addition, we show that their representations contain the information necessary to model behavior on the level of individual subjects. Finally, we demonstrate that finetuning on multiple tasks enables large language models to predict human behavior in a previously unseen task. Taken together, these results suggest that large, pre-trained models can be adapted to become generalist cognitive models, thereby opening up new research directions that could transform cognitive psychology and the behavioral sciences as a whole.

1 Introduction

Large language models are neural networks trained on vast amounts of data to predict the next token for a given text sequence [Brown et al., 2020]. These models display many emergent abilities that were not anticipated by extrapolating the performance of smaller models [Wei et al., 2022]. Their abilities are so impressive and far-reaching that some have argued that they show sparks of general intelligence [Bubeck et al., 2023]. We may currently witness one of the biggest revolutions in artificial intelligence, but the impact of modern language models is felt far beyond, permeating into education [Kasneci et al., 2023], medicine [Li et al., 2023], and the labor market [Eloundou et al., 2023].

In-context learning – the ability to extract information from a context and to use that information to improve the production of subsequent outputs – is one of the defining features of such models. It is through this mechanism that large language models are able to solve a variety of tasks, ranging from translation [Brown et al., 2020] to analogical reasoning [Webb et al., 2022]. Previous work has shown that these models can even successfully navigate when they are placed into classic psychological experiments [Binz and Schulz, 2023, Coda-Forno et al., 2023, Dasgupta et al., 2022, Hagendorff et al., 2022]. To provide just one example, GPT-3 – an autoregressive language model designed by OpenAI [Brown et al., 2020] – outperformed human subjects in a sequential decision-making task that required to balance between exploitative and exploratory actions [Binz and Schulz, 2023].

Even though these models show human-like behavioral characteristics in some situations, this is not always the case. In the sequential decision-making task mentioned above, for instance, GPT-3

Preprint. Under review.

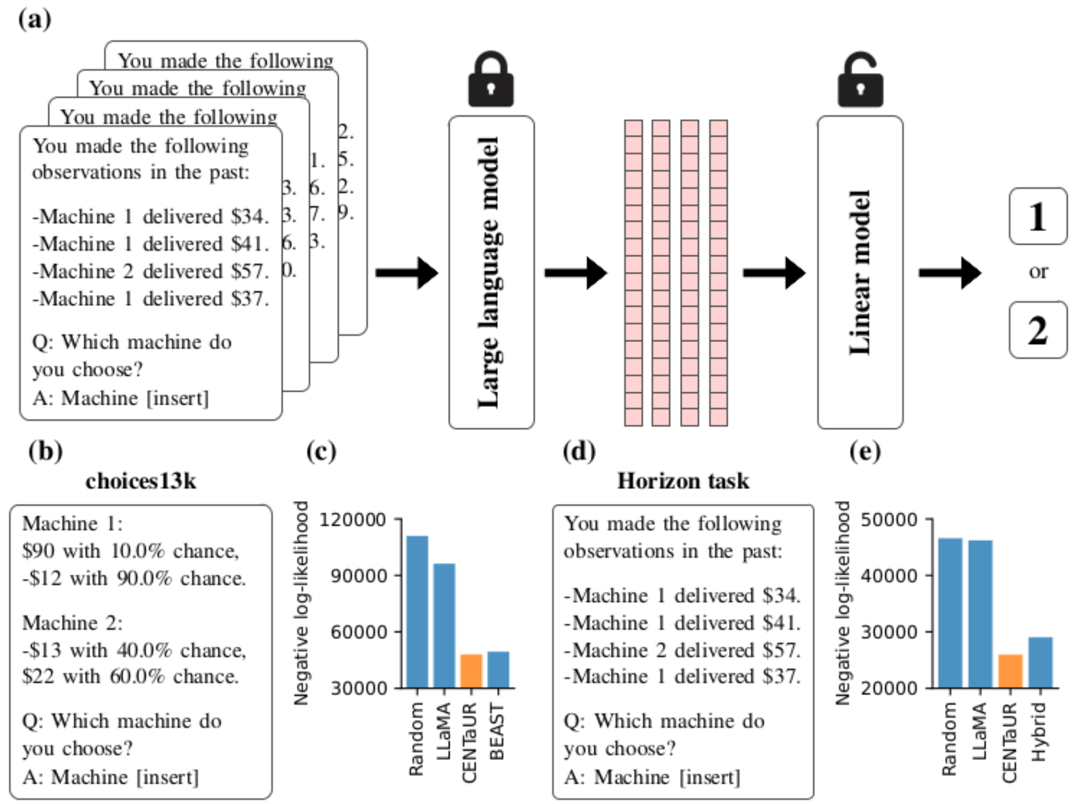

Figure 1: Illustration of our approach and main results. (a) We provided text-based descriptions of psychological experiments to a large language model and extracted the resulting embeddings. We then finetuned a linear layer on top of these embeddings to predict human choices. We refer to the resulting model as CENTaUR. (b) Example prompt for the choices13k data set. (c) Negative log-likelihoods for the choices13k data set. (d) Example prompt for the horizon task. (e) Negative log-likelihoods for the horizon task. Prompts shown in this figure are stylized for readability. Exact prompts can be found in the Supplementary Materials.

relied heavily on exploitative strategies, while people applied a combination of elaborate exploration strategies [Wilson et al., 2014]. Moreover, GPT-3 stopped improving after only a few trials, while people continued learning as the task progressed.

In the present paper, we investigate whether it is possible to fix the behavioral discrepancy between large language models and humans. To do so, we rely on the idea of finetuning on domain-specific data. This approach has been fruitful across a number of areas [Sanh et al., 2019, Ouyang et al., 2022] and eventually led to the creation of the term *foundation models* [Bommasani et al., 2021] – models trained on broad data at scale and adapted to a wide range of downstream tasks. In the context of human cognition, such domain-specific data can be readily accessed by tapping the vast amount of behavioral studies that psychologists have conducted over the last century. We made use of this and extracted data sets for several behavioral paradigms which we then used to finetune a large language model.

We show that this approach can be used to create models that describe human behavior better than traditional cognitive models. We verify this result through extensive model simulations, which confirm that finetuned language models indeed show human-like behavioral characteristics. Furthermore, we find that the embeddings obtained from such models contain the information necessary to capture individual differences. Finally, we highlight that a model finetuned on two tasks is capable of predicting human behavior on a third, hold-out task. Taken together, our work demonstrates that it is possible to turn large language models into cognitive models, thereby opening up completely new opportunities to harvest the power of large language models for building domain-general models of human learning and decision-making.

2 Finetuned language models beat domain-specific models

We started our investigations by testing whether it is possible to capture how people make decisions through finetuning a large language model. For our analyses, we relied on the *Large Language Model Meta AI*, or in short: LLaMA [Touvron et al., 2023]. LLaMA is a family of state-of-the-art foundational large language models (with either 7B, 13B, 33B, or 65B parameters) that were trained on trillions of tokens coming from exclusively publicly available data sets. We focused on the largest of these models – the 65B parameter version – for the analyses in the main text. LLaMA is publicly available, meaning that researchers are provided with complete access to the network architecture including its pre-trained weights. We utilized this feature to extract embeddings for several cognitive tasks and then finetuned a linear layer on top of these embeddings to predict human choices (see Figure 1a for a visualization). We call the resulting class of models CENTaUR, in analogy to the mythical creature that is half human and half ungulate.

We considered two paradigms that have been extensively studied in the human decision-making literature for our initial analyses: *decisions from descriptions* [Kahneman and Tversky, 1972] and *decisions from experience* [Hertwig et al., 2004]. In the former, a decision-maker is asked to choose between one of two hypothetical gambles like the ones shown in Figure 1b. Thus, for both options, there is complete information about outcome probabilities and their respective values. In contrast, the decisions from experience paradigm does not provide such explicit information. Instead, the decision-maker has to learn about outcome probabilities and their respective values from repeated interactions with the task as shown in Figure 1d. Importantly, this modification calls for a change in how an ideal decision-maker should approach such problems: it is not enough to merely exploit currently available knowledge anymore but also crucial to explore options that are unfamiliar [Schulz and Gershman, 2019].

For both these paradigms, we created a data set consisting of embeddings and the corresponding human choices. We obtained embeddings by passing prompts that included all the information that people had access to on a given trial through LLaMA and then extracting the hidden activations of the final layer (see Figure 1b and d for example prompts, and the Supplementary Materials for a more detailed description about the prompt generation procedure). We relied on publicly available data from earlier studies in this process. In the decisions from descriptions setting, we used the choices13k data set [Peterson et al., 2021], which is a large-scale data set consisting of over 13,000 choice problems (all in all, 14,711 participants made over one million choices on these problems). In the decisions from experience setting, we used data from the horizon task [Wilson et al., 2014] and a replication study [Feng et al., 2021], which combined include 60 participants making a total of 67,200 choices.

With these two data sets at hand, we fitted a regularized logistic regression model from the extracted embeddings to human choices. In this section, we restricted ourselves to a joint model for all participants, thereby neglecting potential individual differences (but see one of the following sections for an analysis that allows for individual differences). Model performance was measured through the predictive log-likelihood on hold-out data obtained using a 100-fold cross-validation procedure. We standardized all input features and furthermore applied a nested cross-validation for tuning the hyperparameter that controls the regularization strength. Further details are provided in the Materials and Methods section.

We compared the goodness-of-fit of the resulting models against three baselines: a random guessing model, LLaMA without finetuning (obtained by reading out log-probabilities of the pre-trained model), and a domain-specific model (*Best Estimate and Sampling Tools*, or BEAST, for the choices13k data set [Erev et al., 2017] and a *hybrid model* [Gershman, 2018] that involves a combination of different exploitation and exploration strategies for the horizon task). We found that LLaMA did not capture human behavior well, obtaining a negative log-likelihood (NLL) close to chance-level for the choices13k data set (NLL = 96248.5) and the horizon task (NLL = 46211.4). However, finetuning led to models that captured human behavior better than the domain-specific models under consideration. In the choices13k data set, CENTaUR achieved a negative log-likelihood of 48002.3 while BEAST only achieved a negative log-likelihood of 49448.1 (see Figure 1c). In the horizon task, CENTaUR achieved a negative log-likelihood of 25968.6 while the hybrid model only achieved a negative log-likelihood of 29042.5 (see Figure 1e). Together, these results suggest that the representations extracted from large language models are rich enough to attain state-of-the-art results for modeling human decision-making.